3.1.5 The functioning of the heart plays a central role in the circulation of blood and relates to the level of activity of an individual. Heart disease may be linked to factors affecting lifestyle.

Heart structure and function	The gross structure of the human heart and its associated blood vessels in relation to function. Pressure and volume changes and associated valve movements during the cardiac cycle. Myogenic stimulation of the heart and transmission of a subsequent wave of electrical activity. Roles of the sinoatrial node (SAN), atrioventricular node (AVN) and bundle of His. Cardiac output as the product of heart rate and stroke volume. **Candidates should be able to** analyse and interpret data relating to pressure and volume changes during the cardiac cycle.
The biological basis of heart disease	Atheroma as the presence of fatty material within the walls of arteries. The link between atheroma and the increased risk of aneurysm and thrombosis. Myocardial infarction and its cause in terms of an interruption to the blood flow to heart muscle. Risk factors associated with coronary heart disease: diet, blood cholesterol, cigarette smoking and high blood pressure. **Candidates should be able to** describe and explain data relating to the relationship between specific risk factors and the incidence of coronary heart disease.